

A large-scale comparison of divergent creativity in humans and large language models

Received: 13 November 2024

Accepted: 25 September 2025

Published online: 23 December 2025

 Check for updates

Dawei Wang^{1,2}✉, Difang Huang^{3,4}, Haipeng Shen¹ & Brian Uzzi^{2,5,6}✉

Human–machine partnerships are increasingly used to address grand societal challenges, yet knowledge of the comparative strengths of humans and machines to innovate is nascent. Here we compare the ability of humans ($N = 9,198$) and large language models (LLMs, $N = 215,542$ observations) to generate novel ideas in an established creativity task. We present three key results. First, human creativity on average is slightly higher than that of LLMs. Second, creativity differences are pronounced at the extremes of the distribution, with humans exhibiting greater variability and higher levels of creativity in the right-hand tail of the distribution. Third, attempts to increase the creativity of LLMs through instructing LLMs to take on genius personas or different demographic roles lifted performance up to a threshold beyond which the output became opposite real-life patterns, whereas strategic prompt-engineering efforts yielded mixed to negative results. We discuss the implications of our findings for human–machine collaboration and problem solving.

Einstein's creative genius reimagined the universe through the concept of spacetime and opened a gateway to far-reaching advances in communication, transportation, energy and medical science. Beyond its instrumental role in problem solving, creativity serves as a fundamental source of inspiration across science, business and the arts^{1,2} and is an innate human drive as observed in infants, who are predisposed to spend substantial portions of their development recombining familiar objects into novel configurations³.

Recent advances in artificial intelligence (AI) and natural language processing (NLP), particularly in large language models (LLMs) such as the generative pre-trained transformer (GPT), have created opportunities for augmenting human creativity. Yet our understanding of the comparative creative abilities of humans and LLMs remains nascent^{4–8}. A few studies have begun to examine LLM performance on tests designed to measure human creativity. The Alternative Uses Test,

which assesses creativity by evaluating the ability to generate unconventional uses for everyday objects⁹, suggests that LLMs can match or even surpass human performance^{10,11}. Yet fundamental questions remain regarding the inherent limitations of LLM creative diversity compared with humans^{12–14}. For instance, the emergence of specialized reasoning models such as DeepSeek-R1 (ref. 15) raises new questions about whether these enhanced systems have outperformed earlier generations of LLMs and whether their advancements have improved their capacity to exceed human creativity^{16,17}.

Our study advances the literature in three ways. First, in contrast to prior work that predominantly relied on evaluations by human judges, we employed the Divergent Association Task (DAT)—a widely-used and more scalable and replicable measure—to assess creativity in both humans and LLMs^{18,19}. The DAT is a validated test designed to measure a person's divergent creativity. It requires participants to name ten nouns

¹Faculty of Business and Economics, University of Hong Kong, Hong Kong, China. ²Northwestern Institute on Complex Systems, Northwestern University, Evanston, IL, USA. ³Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China. ⁴School of Economics and Management, University of Chinese Academy of Sciences, Beijing, China. ⁵Department of Management and Organizations, Kellogg School of Management, Northwestern University, Evanston, IL, USA. ⁶Ryan Institute of Complexity, Kellogg School of Management, Northwestern University, Evanston, IL, USA. ✉e-mail: dwdw@hku.hk; uzzi@northwestern.edu

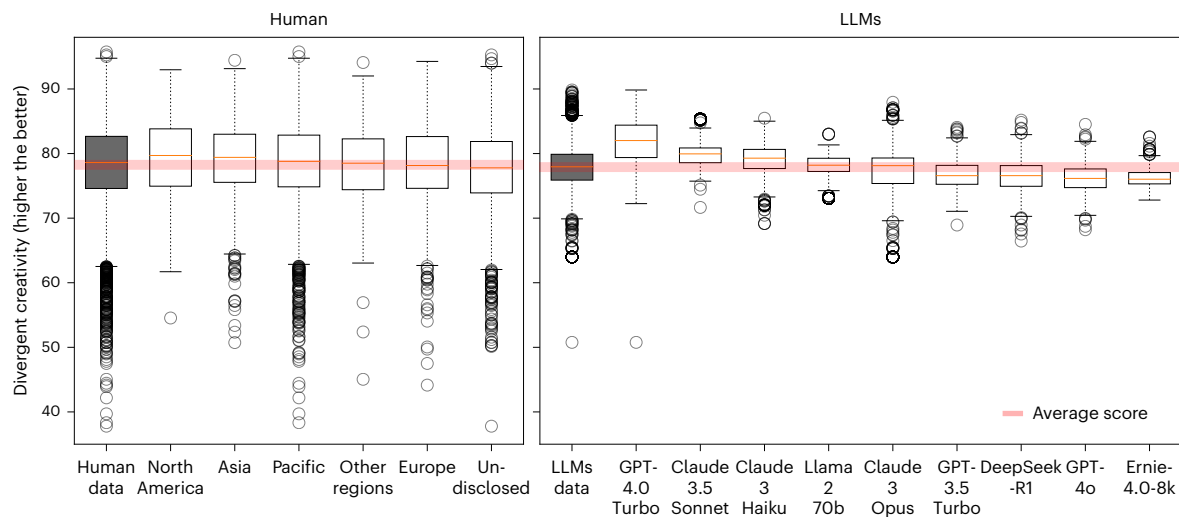


Fig. 1 | Comparison of the divergent creativity scores between humans and LLMs. A comparison of divergent creativity scores from humans (left, by region) and LLMs (right). The box plots show medians, interquartile ranges and outliers; whiskers extend to $1.5 \times$ the interquartile range. The red line marks the overall

mean. Total sample size for humans = 9,198 (sample sizes range: 139–4,922); for LLMs = 6,746 (sample sizes range: 746–750). For details, see Supplementary Information.

that differ as much as possible from each other without the aid of external cues or word sources. Each noun is paired with each other noun, and the level of difference between the nouns in all pairs is quantified by an algorithm that computes the semantic difference between words. For example, the semantic difference between king and queen is the same as for man and woman, but less than for man and boy and more than for king and turnip²⁰. This form of creativity is called divergent creativity and has been shown to be a root cause of human cognition in finding novel solutions to open-ended problems^{21,22}. Second, our computation of an algorithmic score meant that all results by humans and LLMs were transparently quantified^{23–27} and replicable²⁸. Note that we excluded tests like the Torrance Tests of Creative Thinking and Remote Associates Task because the length and complex scoring of the Torrance Tests of Creative Thinking presented feasibility and comparability issues for large-scale evaluations of creativity between humans and LLMs and the Remote Associates Task measures convergent, not divergent thinking (our study's focus). Third, our study draws from a culturally diverse human participant pool spanning multiple continents and evaluates a range of LLMs, including the recently released reasoning model DeepSeek-R1 (ref. 15). By incorporating this broad spectrum of participants and models, we aim to deepen our understanding of the comparative creative abilities of humans and LLMs.

Our findings reveal differences between human and LLM divergent creativity. While the average performance of humans is only slightly higher than that of LLMs, humans exhibit greater variance, with exceptional performances at the upper end of the distribution and correspondingly lower performance at the lower end, whereas LLMs display more uniform performance. Additionally, humans generate more varied responses, whereas LLM outputs tend to be relatively homogeneous. Testing boundary conditions revealed consistent patterns: increasing model randomness initially enhanced LLM creativity but ultimately resulted in incoherent outputs²⁹. Contrary to common assumptions³⁰, LLM performance declined when prompted to adopt creative personas and their outputs diverged from actual human patterns when instructed to assume specific demographic characteristics.

Our findings provide insights into the future of human–machine collaboration and address questions about whether LLMs will supersede human cognitive capabilities. The results present a nuanced perspective: while LLMs can match the average human creative performance, highlighting their potential to enhance baseline creativity and

operational efficiency, they do not replicate the exceptional creative abilities demonstrated by high-performing humans. This distinction underscores the unique value of human creativity, which is shaped by understanding, emotions and lived experiences. These findings suggest that rather than replacing humans, LLMs may be best positioned as tools to augment human creative processes, particularly by supporting routine creative tasks while leaving expert-level creativity to human specialists.

Results

Study 1A: baseline

Figure 1 and Table 1 compare the baseline, average divergent creativity between humans and LLMs in Study 1A. Figure 1 (left) presents human performance across geographic regions. The mean creativity score for all humans in our sample is 78.26. North American participants recorded an average score of 79.28, Asia averaged 78.82, the Pacific regions averaged 78.53, Europe averaged 77.96, other regions, such as African regions, averaged 78.03 and undisclosed locations averaged 77.52 (for details on the human data, see Supplementary Note 1; for an ANOVA analysis on the human averages between different regions, see online supplementary, <https://osf.io/a9v2t>). Figure 1 (right) depicts the performance of nine models in our LLM sample (for an analysis of variance (ANOVA) analysis on the LLM averages between different models, see online supplementary, <https://osf.io/a9v2t>). OpenAI's GPT-4.0 Turbo achieved the highest mean score among LLMs at 81.78 ($t = 35.52$, $P < 0.001$, 95% confidence interval (CI) 4.13–4.61, Cohen's $d = 1.38$), exceeding the human average in our dataset (Supplementary Table 5). Claude 3.5 Sonnet and Claude 3 Haiku followed with mean scores of 80.01 ($t = 18.14$, $P < 0.001$, 95% CI 2.12–2.63, Cohen's $d = 0.70$) and 79.06 ($t = 9.80$, $P < 0.001$, 95% CI 1.04–1.56, Cohen's $d = 0.38$), respectively. The remaining models, including Llama 2 70b, Claude 3 Opus, GPT-3.5 Turbo, GPT-4o, DeepSeek-R1 and Ernie-4.0-8k, produced mean divergent creativity scores below the human average (for example, Ernie-4.0-8k: $t = 22.29$, $P < 0.001$, Cohen's $d = 0.33$; Supplementary Table 7). These results align with previous findings comparing human and LLM creative performance, reinforcing that humans, particularly experts, outperform LLMs in complex creative thinking tasks^{31,32}. A two-sided t -test revealed a notable difference between all human and LLM responses in Study 1A ($t = 3.99$, $P < 0.001$, Cohen's $d = 0.06$; Table 1). Across all models, both the average and top-performing human participants significantly outperformed LLMs

Table 1 | Statistical test of average human versus LLM divergent creativity performance

Test	Human	LLMs	Parametric test statistic	Parametric test <i>P</i> value	Parametric test 95% CI	Parametric test Cohen's <i>d</i>	Non-parametric test statistic	Non-parametric test <i>P</i> value
Score mean (entire sample)	78.26	77.90	3.99***	<0.001	0.18–0.53	0.06	3,3458,265.00***	<0.001
Score variance (entire sample)	45.26	11.95	3,033.37***	<0.001	–	–	1,760.58***	<0.001
Score mean (90th percentile)	88.80	84.29	48.84***	<0.001	4.32–4.68	2.48	601,842.00***	<0.001
Score variance (90th percentile)	3.87	2.54	33.56***	<0.001	–	–	24.70***	<0.001
Mean of valid words (100 observations)	950.34	999.87	205.30***	<0.001	49.06–50.01	9.18	1,000,000.00***	<0.001
Variance of valid words (100 observations)	58.03	0.13	4,734.66***	<0.001	–	–	1,641.08***	<0.001
Mean of unique words (100 observations)	577.00	181.05	552.46***	<0.001	394.54–397.35	24.71	1,000,000.00***	<0.001
Variance of unique words (100 observations)	403.29	109.85	395.42***	<0.001	–	–	353.26***	<0.001

This table compares human and LLM performance on the divergent creativity task using parametric and non-parametric tests. Reported are *t* values, *P* values, Cohen's *d* and 95% CIs. All tests were two sided without correction for multiple comparisons. **P* < 0.05, ***P* < 0.01, ****P* < 0.001.

on the divergent creativity task (for example, GPT-4.0 Turbo: $t = 48.71$, $P < 0.001$, Cohen's $d = 2.53$; Ernie-4.0-8k: $t = 137.65$, $P < 0.001$, Cohen's $d = 6.71$; Supplementary Tables 7 and 8).

Whereas first-order statistics suggest that some LLMs, such as GPT-4.0 Turbo, may outperform humans, second-order statistics reveal significant differences in divergent creativity performance between humans and LLMs and that humans may exhibit exceptional creativity at the upper end of the distribution. Specifically, humans demonstrated greater variance in their divergent creativity scores compared with LLMs (Table 1), with both parametric (Bartlett's test 3,033.37, $P < 0.001$) and non-parametric (Levene's test 1,760.58, $P < 0.001$) analyses confirming statistical significance. To quantify the exceptional creative performance of top-performing humans, we compared the top tenth percentile of human participants with the highest-performing LLM in our sample, GPT-4.0 Turbo. The results showed that high-performing humans significantly outperformed GPT-4.0 Turbo's top outputs (Supplementary Table 5), with both parametric and non-parametric tests yielding statistical significance ($t = 6.73$, $P < 0.001$, Cohen's $d = 0.81$, 95% CI 1.10–2.00). A parallel comparison with DeepSeek-R1 produced similar results ($t = 34.46$, $P < 0.001$, Cohen's $d = 0.27$, 95% CI 7.50–8.41; Supplementary Table 6).

Another metric used to evaluate creative capability was the count of unique words generated by humans and LLMs, assessed through bootstrapping observations and comparing randomly drawn words from a bag of words generated by each group^{33–35}. We found that while LLMs generated a higher number of valid words than humans, they also repeated words significantly more often, resulting in a lower proportion of unique words compared with human responses (mean of unique words: $t = 552.46$, $P < 0.001$, Cohen's $d = 24.71$, 95% CI 394.54–397.35; Table 1). For example, an LLM might generate a response such as 'apple, cloud, chair,...' and, in an independent chat session, produce 'chair, apple, cloud,...' (that is, exactly the same words but in different order) demonstrating a tendency to recycle words rather than to generate truly novel responses (Supplementary Figs. 3 and 4).

Study 1A revealed nuanced findings regarding the divergent creativity performance of humans and LLMs. On average, human performance on the divergent creativity task was slightly higher than that of the LLMs in our sample. However, compared with LLMs, humans exhibited significantly greater variability in their divergent creativity scores and, on average, generated a broader repertoire of unique

outputs. Taken together, these findings reinforce previously identified limitations in LLMs, which tend to produce more uniform responses to similar prompts³⁶.

Study 1B: temperature

In Study 1B, we used the same baseline prompt as in Study 1A but manipulated the 'creativity' parameter in LLMs. This parameter, often referred to as the 'temperature' setting, adjusts the randomness of word generation and is commonly linked to model creativity (Supplementary Fig. 5). In contrast, humans typically exhibit systematic patterns of controlled variability in cognitive tasks, balancing novelty with meaningful conceptual relationships^{37,38}. We tested temperature settings ranging from 0.1 to 1.0, measuring the divergent creativity scores of each model across different temperatures and comparing them with the baseline scores of humans and LLMs in Study 1A (temperature of 0.5).

Figure 2 illustrates an overall increase in divergent creativity scores across the nine LLMs as the temperature parameter rises, with ANOVA analysis confirming statistical significance ($F(9, 66,199) = 157.45$, $P < 0.001$, $\eta_p^2 = 0.02$; Supplementary Table 12). As the temperature increased from 0.1 to 1.0, the number and variance of unique words generated by LLMs also rose significantly (mean of unique words: $t = 308.81$, $P < 0.001$, Cohen's $d = 13.81$, 95% CI 215.23–217.98; Supplementary Table 9). Among the models tested, GPT-based models exhibited the most pronounced increase in divergent creativity scores as temperature increased (Fig. 2).

However, despite the improvements from temperature tuning, the top human sample continued to significantly outperform LLMs (for example, $t = 159.12$, $P < 0.001$, Cohen's $d = 4.76$ at temperature 0.1; $t = 116.22$, $P < 0.001$, Cohen's $d = 2.50$ at temperature 1.0; Supplementary Table 14). Supplementary Table 13 reports *t*-tests comparing average human and LLM performance across temperature settings from 0.1 to 1.0. Most pairwise comparisons yielded significant differences ($P < 0.001$, for example, $t = 10.68$, $P < 0.001$, Cohen's $d = 0.15$ at temperature 0.1; $t = -8.04$, $P < 0.001$, Cohen's $d = 0.12$ at temperature 1.0). Moreover, while higher temperatures increased LLM DAT scores, the quality of generated words deteriorated (Supplementary Figs. 8 and 9). As temperature increased, LLMs produced an increasing number of nonsensical words, including letter combinations from different languages, mixtures of letters and symbols or words that do not appear in any dictionary (Fig. 2). Similarly, DeepSeek-R1's 'chain-of-thought process' failed frequently at higher

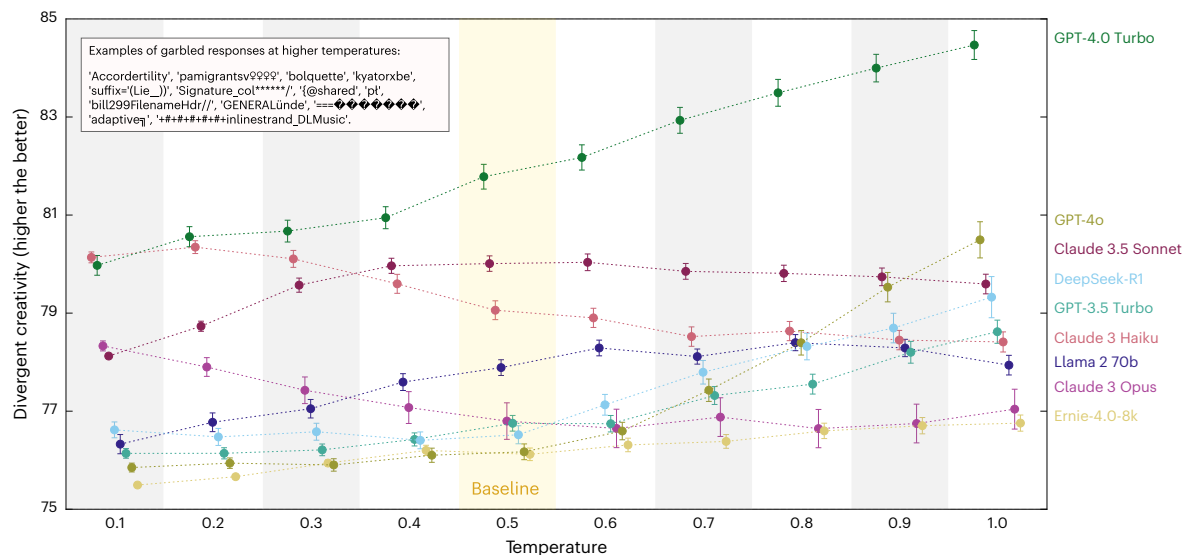


Fig. 2 | Comparison of divergent creativity scores across different temperature values for LLMs. The divergent creativity scores of nine LLMs across temperature values ranging from 0.1 to 1.0, with baseline set at 0.5. Temperature controls output randomness: lower values yield predictable responses; higher values increase word variability. Each dot represents a model's mean score at a given

temperature, with 95% CIs. The inset shows examples of garbled responses at high temperatures. Total sample size for baseline prompt = 6,746 (sample sizes: 746–750); for engineered prompt sample size = 66,209 (sample sizes: 387–750). For details, see Supplementary Information.

temperatures (for details on DeepSeek-R1 and other LLM-specific changes related to temperature adjustments, see Supplementary Note 4). Overall, Study 1B finds that increasing model temperature generally enhances LLM creativity but does not ensure exceptional creativity at the level of top-performing humans, nor does it guarantee that LLMs can discern semantic meaning—a crucial aspect of creativity that links creative thinking to usefulness³⁹.

Study 2A: perspective-taking prompts

Building on the baseline prompts in studies 1A and 1B, studies 2A, 2B and 2C examined how modifications to LLM instructions influence their responses^{40,41}. In Study 2A, we directed the models to adopt five distinct perspectives, each emphasizing different aspects of the task^{41,42}: task related (instructing the LLM to rely on its memory of DAT from its training data), creativity related (explicitly telling the LLM to be creative), outcome related (specifying the desired outcome), imagination related (asking the LLM to use its own imagination to generate responses) and a combination of every strategy above (combining creativity, imagination and outcome-driven instructions). However, note that a fundamental limitation of prompt engineering is the difficulty of verifying whether human and LLM interpretations of the instructions align, given the black-box nature of LLMs^{43–45}. Finally, we compared the results of these modified prompts to the baseline human and LLM performance in Study 1A.

As shown in Fig. 3, most prompt-engineering approaches improved divergent creativity scores beyond both human and LLM baseline performance (detailed analyses are reported in Supplementary Note 2). Statistical analyses in Supplementary Tables 15 and 16 confirm that these improvements are statistically significant ($P < 0.001$). Compared with baseline LLMs, the mean divergent creativity score for perspective-taking prompts was significantly higher ($t = 33.25$, $P < 0.001$, Cohen's $d = 0.42$, 95% CI 1.72–1.93; Supplementary Table 15). Perspective-taking prompts also significantly outperformed human responses ($t = 27.99$, $P < 0.001$, Cohen's $d = 0.31$, 95% CI 1.37–1.57; Supplementary Table 16). Among the models, GPT-4.0 Turbo demonstrated the strongest positive response across all engineered instructions, with significant improvements observed under perspective prompts (for example, 'creativity': $t = -17.73$, $P < 0.001$,

Cohen's $d = 0.79$; 'everything': $t = -13.61$, $P < 0.001$, Cohen's $d = 0.64$; Supplementary Table 19), whereas other LLMs exhibited varying levels of responsiveness (Fig. 3). Unexpectedly, Claude 3 Opus did not show statistically significant improvement when explicitly prompted to generate diverse words in the 'imagination' condition ($t = -0.12$, $P = 0.900$, Cohen's $d = 0.01$; Supplementary Table 20).

Examining second-order statistics revealed that while prompt engineering improved average performance compared with Study 1A, it did not enable models to surpass humans in linguistic variance or unique word generation (Supplementary Note 1). Even under the perspective-taking condition, humans continued to outperform LLMs. Even under the perspective-taking condition, humans continued to show distinctive advantages over LLMs. As presented in Supplementary Table 16, humans produced significantly more unique words ($t = 429.74$, $P < 0.001$, Cohen's $d = 19.22$, 95% CI 323.58–326.54) and exhibited greater variance in valid word usage ($t = 2,663.70$, $P < 0.001$) compared with LLMs. However, when performance was analysed at the individual prompt level, a more nuanced pattern emerged. Supplementary Table 17 shows that LLMs outperformed the average human across all perspective prompts, including 'creativity' ($t = -19.57$, $P < 0.001$, Cohen's $d = 0.29$) and 'imagination' ($t = -18.76$, $P < 0.001$, Cohen's $d = 0.28$). This trend was reversed when focusing on the top-performing human participants. As shown in Supplementary Table 18, top humans consistently and substantially outperformed LLMs across all prompts (for example, 'task recall': $t = 131.00$, $P < 0.001$, Cohen's $d = 2.58$; 'imagination': $t = 123.13$, $P < 0.001$, Cohen's $d = 2.30$), highlighting the enduring advantage of human excellence, even under optimized prompting conditions. These findings suggest that while prompt engineering can enhance performance, LLMs still fall short of human creativity, particularly at the highest levels of performance.

Study 2B: creative genius prompts

In Study 2B, we examined whether LLM creative performance could be enhanced by prompting LLMs to adopt the personas of historically renowned creative figures (for a detailed description of these personas, see Supplementary Note 5). We instructed the LLMs to assume the perspectives of eight distinguished individuals across diverse creative

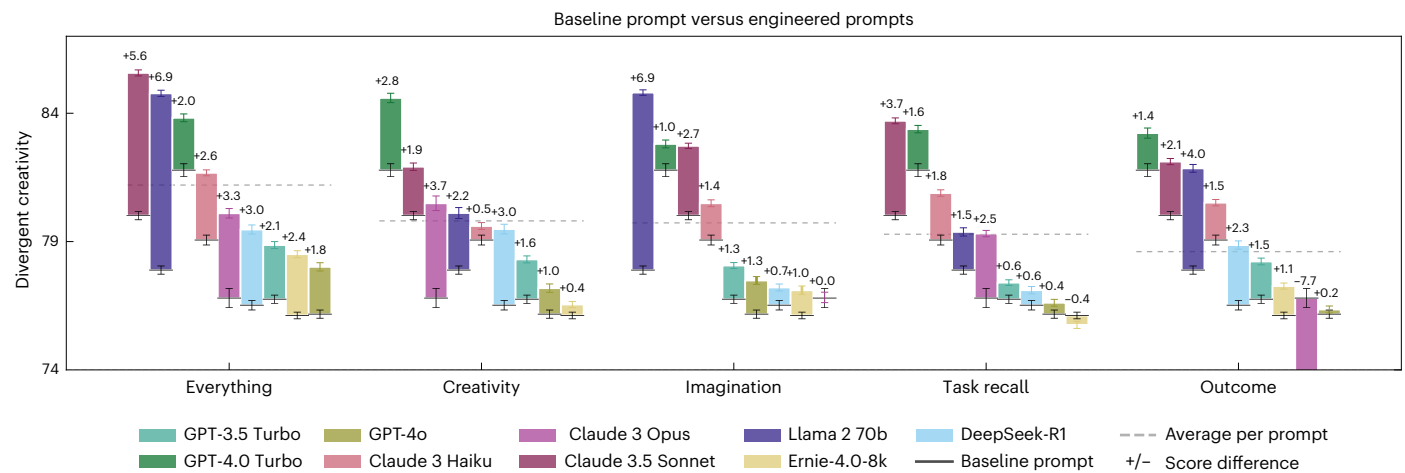


Fig. 3 | Comparison of divergent creativity scores across different perspective prompts for LLMs. A bar graph showing the average divergent creativity scores of all LLMs across five perspective prompts. The bars represent mean DAT scores with 95% CIs. Dashed lines indicate the overall average across all engineered

prompts. Total sample size = 74,140; baseline prompt sample size = 6,746 (sample sizes: 746–750); engineered prompt sample size = 67,394 (sample sizes: 1,475–1,500). For details, see Supplementary Information.

domains: artists (Georgia O’Keeffe and Leonardo da Vinci), scientists (Marie Curie and Albert Einstein), entrepreneurs (Oprah Winfrey and Steve Jobs) and writers (William Shakespeare and Noam Chomsky). We then measured their divergent creativity scores and compared them with the baseline performance of both humans and LLMs from Study 1A.

As depicted in Fig. 4, LLMs’ divergent creativity scores generally declined when adopting the personas of exceptionally creative individuals (detailed analyses are reported in Supplementary Note 2). This counterintuitive effect is further supported by Supplementary Table 23, which presents pairwise comparisons between all human participants and LLMs prompted with different celebrity personas. LLMs were consistently outperformed by the general human population across all celebrity prompts (for example, $t = 41.13$, $P < 0.001$, Cohen’s $d = 0.63$ for O’Keeffe), their performance advantage was reduced compared with baseline prompts. Notably, Supplementary Table 24 shows that when compared against top-performing humans, LLMs fell significantly short across all celebrity conditions, with large effect sizes (for example, $t = 170.71$, $P < 0.001$, Cohen’s $d = 4.55$ for Shakespeare). Overall, these findings indicate that when LLMs approach the creativity task from the perspective of creative masterminds, their performance deteriorates.

Study 2C: demographic group prompts

In Study 2C, we examined LLMs’ creative performance when instructed to adopt different demographic perspectives represented in our human DAT data. Specifically, we focused on two gender categories (female and male) and two age groups (old and young). We then compared these results with both human performance and baseline LLM scores from Study 1A.

The changes in LLMs’ divergent creativity scores when given demographic-specific prompts relative to their baseline performance in Study 1A are illustrated in Fig. 5, top. As presented in Supplementary Table 29, gender-based prompts such as ‘female’ ($t = 18.38$, $P < 0.001$, Cohen’s $d = 0.35$) and ‘male’ ($t = 3.44$, $P < 0.001$, Cohen’s $d = 0.06$) led to significantly lower creativity scores compared with the baseline. Age-based ‘young’ prompt yielded a small but significant improvement ($t = -2.12$, $P = 0.030$, Cohen’s $d = 0.04$), whereas the ‘old’ prompt resulted in a substantial enhancement in performance ($t = -15.83$, $P < 0.001$, Cohen’s $d = 0.32$).

There are intriguing differences between LLM responses to demographic prompts and actual human demographic group performance (Fig. 5, bottom, and Supplementary Table 30). A two-way ANOVA (Supplementary Table 27) confirmed significant main effects of type

($F(1, 15,938) = 269.82$, $P < 0.001$, $\eta_p^2 = 0.02$) and gender ($F(1, 15,938) = 15.26$, $P < 0.001$, $\eta_p^2 < 0.01$) on divergent creativity scores, as well as a significant interaction between group and gender ($F(1, 15,938) = 108.01$, $P < 0.001$, $\eta_p^2 = 0.01$). These results indicate that LLMs not only differ from humans overall but also fail to replicate the gender-related patterns observed in human responses. However, when benchmarked against top-performing humans (Supplementary Table 31), LLMs fell significantly short across all demographic prompts, with large effect sizes (for example, $t = 145.88$, $P < 0.001$, Cohen’s $d = 3.70$ for ‘female’; $t = 143.74$, $P < 0.001$, Cohen’s $d = 4.14$ for ‘male’; and $t = 103.78$, $P < 0.001$, Cohen’s $d = 2.33$ for ‘old’).

Discussion

This study conducted a large-scale comparative evaluation of human and LLM performance on a generative task requiring divergent creativity. Our findings suggest that while humans achieved modestly higher scores than LLMs in the divergent creativity task across the broader sample, the differences between human and LLM creativity are more nuanced. Collectively, human participants exhibited greater linguistic diversity, characterized by both higher variance in task performance and a broader range of vocabulary usage. Further tests, designed to examine LLM performance under specific conditions, reinforced these main findings. We manipulated the temperature, or ‘creativity’ parameter, of LLMs, which was expected to enhance their creativity scores. While higher temperature initially increased divergent creativity, performance deteriorated as LLMs began generating garbled responses when the parameter was pushed to its limit. Additionally, we conducted 17 prompt-engineering trials, categorizing prompting strategies into three distinct types: perspective taking (focusing on specific task aspects), persona taking (adopting the viewpoints of historically creative figures) and demographic taking (simulating responses across different gender and age groups). The results showed that LLMs responded most effectively to perspective-taking prompts that directed attention to specific task components. In contrast, persona-taking prompts consistently resulted in suboptimal performance. Most notably, demographic-taking prompts produced response patterns that inversely correlated with the performance trends observed in the corresponding human demographic groups in our sample.

Our study contributes to theoretical understanding^{46–48} of how humans and LLMs may differentially contribute to creative knowledge and idea generation. Our findings consistently show that LLMs

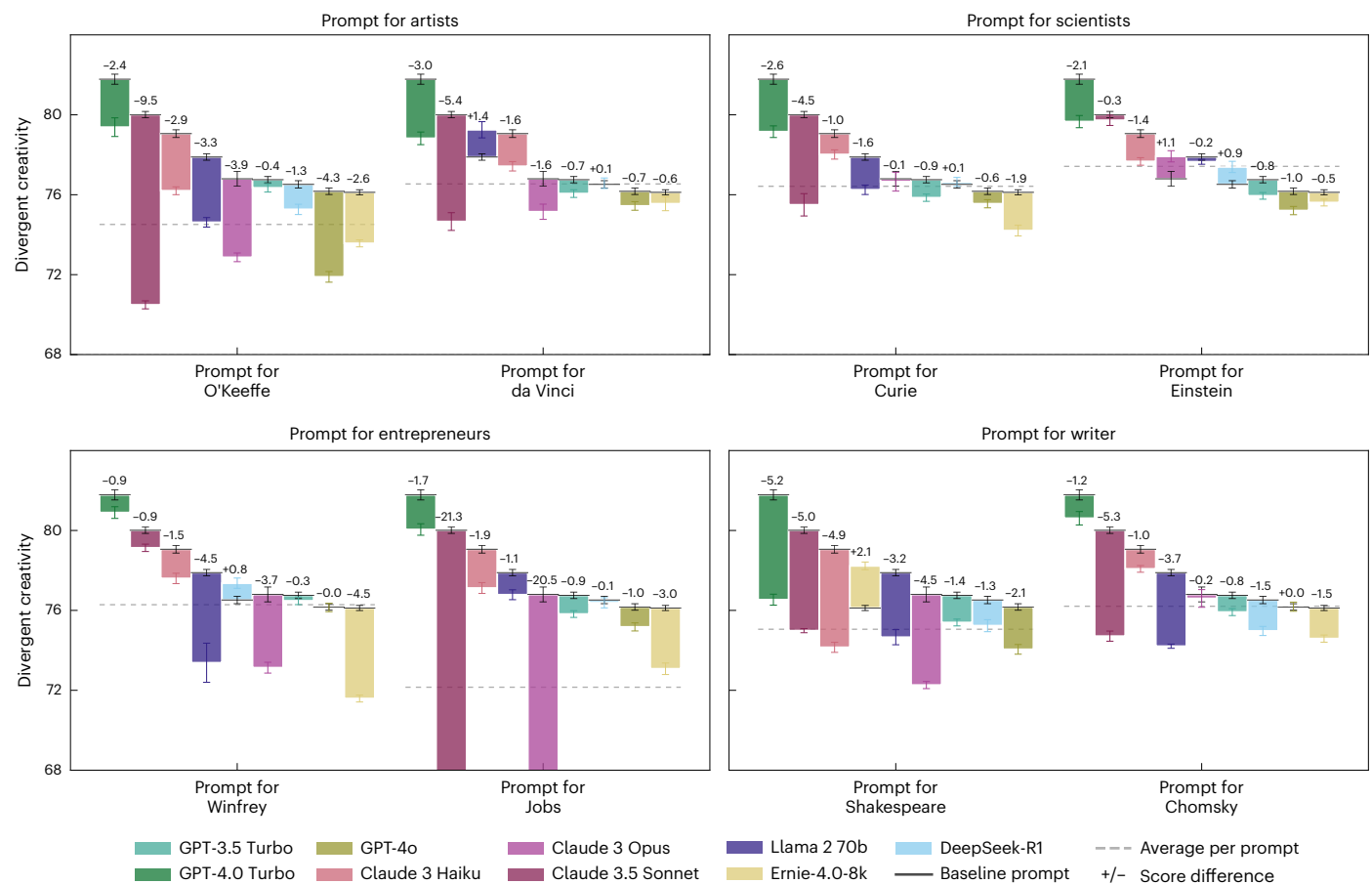


Fig. 4 | Comparison of divergent creativity scores across different celebrity prompts for LLMs. A bar graph illustrating the average DAT scores of all LLMs across eight celebrity viewpoint-taking prompts: artists (O'Keeffe and da Vinci), scientists (Curie and Einstein), entrepreneurs (Winfrey and Jobs) and writers (Shakespeare and Chomsky). The bars represent average divergent thinking

scores for each engineered prompt relative to the baseline, with 95% CIs. Total sample size = 42,555; baseline prompt sample size = 6,746 (sample sizes: 746–750); engineered prompt sample size = 35,809 (sample sizes: 352–500). For details, see Supplementary Information.

lack the full variability and richness of human cognition, a limitation that may stem from factors such as their autoregressive design^{49–51} and the constraints of reinforcement learning with human feedback. While reinforcement learning with human feedback ensures safe and reliable outputs, it may also suppress content diversity and creative expression^{49–54}. These limitations highlight potential risks associated with over-reliance on LLMs for creative tasks as it could inhibit innovative breakthroughs in society. That said, LLMs demonstrate clear utility in generating instantaneous baseline creative outputs, as demonstrated by the consistently higher number of valid words generated compared to humans in our studies, offering initial inspiration and providing foundational ideas from which human creativity can expand. We encourage future research on human–machine augmentation to explore whether baseline responses generated by LLMs can enhance the productivity of creative individuals.

Our findings highlight inherent differences in how humans and LLMs approach creativity. Human creativity emerges from a complex interplay of lived experiences and emotional contexts grounded in understanding and comprehension of the topic. This process may be linked to broader vocabulary knowledge and reading experience, which facilitate intuitive leaps in idea generation. While these mechanisms are difficult to test directly, LLMs, by contrast, probably rely on statistical manipulation and pattern recognition derived from their training data (see additional potential mechanism tests in Supplementary Note 3). However, in our study, participants were only asked to generate ten words, a task that may challenge average human cognition

as it progresses. If the task required a substantially larger number of words, human performance might begin to decline, whereas LLMs, with their vast parameter counts and extensive training data, could offer supplemental support.

These findings suggest future directions for human–machine creative collaboration. For example, consider two practical applications: in marketing, human professionals might generate initial creative concepts for product advertisements, while LLMs systematically explore variations and extensions of these foundational ideas. Similarly, in creative writing, human authors could establish core narrative premises, with LLMs assisting in generating diverse plot possibilities within established narrative frameworks. These complementary relationships leverage the distinct advantages of both systems—human intuitive understanding and contextual awareness combined with AI-driven systematic exploration. We encourage future research to further investigate the differences in how humans and LLMs approach creative tasks.

There are several implications for the future of work and potential partnerships between humans and LLMs^{4,55}. The evolution of labour arrangements has historically been driven by economic efficiency and economies of scale. Our results indicate that LLMs can achieve average levels of creativity, suggesting the potential for LLMs to replace entry-level or even mid-level creative tasks. However, optimizing human–machine partnerships requires more nuanced considerations than simply combining their respective strengths. Integrating these partnerships into organizational structures demands careful strategic planning. While LLMs offer efficiency and scalability, recent evidence

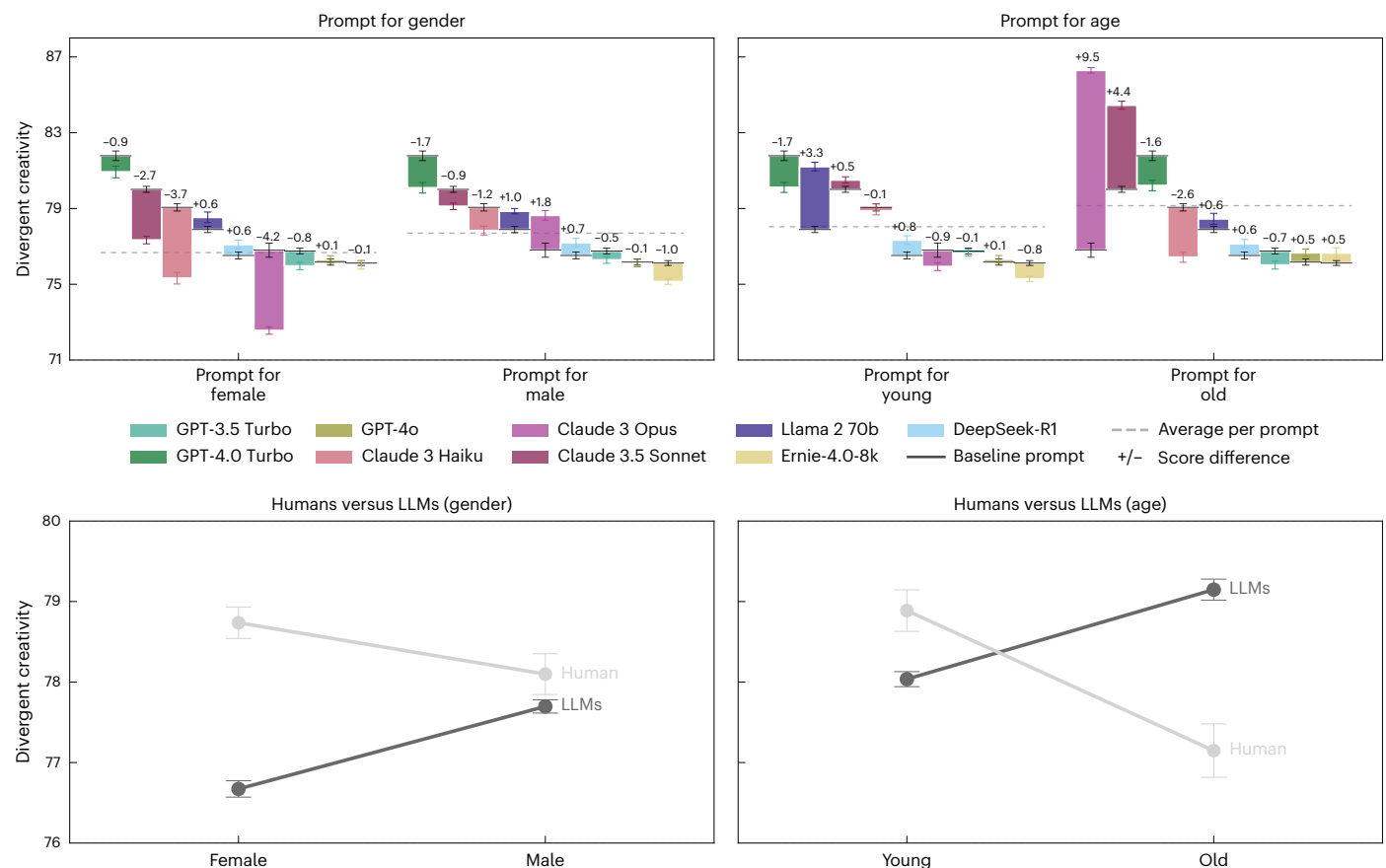


Fig. 5 | Comparison of divergent creativity scores across different demographic prompts for LLMs. Top: the mean DAT scores for nine LLMs in response to gender-based (left) and age-based (right) perspective prompts compared with their respective baselines. Bottom: a comparison of human and LLM performance across gender (female versus male) and age (young versus old). The bars indicate mean divergent creativity scores, with error bars reflecting

95% CIs. The dashed lines denote the average across all LLMs. Total sample size for baseline prompt = 6,746 (sample sizes: 746–750); for engineered prompts sample size = 17,978 (sample sizes: 491–500). Human sample: total sample size = 7,970 (sample sizes: 1,821–4,032); for engineered prompts sample size = 17,978 (sample sizes: 4,491–4,496). For details, see Supplementary Information.

suggests that human–machine collaboration can underperform traditional human teamwork, particularly when task complexity and division of labour are not strategically structured^{56,57}. This highlights the need for thoughtful design in implementing human–machine collaboration, ensuring that complementary strengths are effectively leveraged while minimizing potential inefficiencies or process losses.

Our finding that LLMs exhibit their own form of process losses suggests the need for caution before concluding that LLMs can fully supplant human creativity. Study 1B's manipulation of LLM temperature settings revealed that these models may rely more on statistical randomness than genuine semantic comprehension. This limitation is most evident in our observation that the top percentile of human performers consistently outperformed all LLM performance metrics. Additionally, the superior linguistic diversity demonstrated by human participants, reflected in greater variance in both divergent creativity performance and vocabulary usage, further reinforces this distinction. These results collectively indicate that current LLM technology has not yet mastered high-level creative tasks. However, they also suggest that optimal creative value may arise from configurations in which human–LLM teams effectively mitigate the process limitations inherent in their collaboration. Future research should continue to explore this experimental space and investigate the complex interactions between teams and LLMs.

This study has several limitations. While our sample size was substantial, our data may not fully capture the entire spectrum of human creativity or all possible LLM configurations. That said, the primary

contribution of our study lies in the human–LLM comparison rather than within-human or within-LLM comparisons. Another limitation is our reliance on the DAT, which, while valuable for measuring divergent creativity, captures only one dimension of creative ability and provides a simplified, decontextualized measure that may not fully reflect the complexity of real-world creative processes^{58,59}. As highlighted in the Creativity Diamond framework⁶⁰, various creativity tools such as morphological analysis, SCAMPER and analogical reasoning serve specific purposes in the divergent and convergent phases of creative thinking. Future studies could systematically examine how these methods interact with baseline DAT performance to enhance creative output across different contexts. Building on this potential, recent work has demonstrated promising approaches for augmenting human creativity through computational support, such as TRIZ-GPT⁶¹, which combines TRIZ problem-solving methodology with large language models, and Design-Fusion⁶², which implements generative enrichment models for conceptual design. These advances suggest valuable directions for future research that explores how individuals with varying divergent thinking abilities might benefit from creativity upskilling methods as well as personalized AI-augmented creativity support systems that enhance human creativity and innovativeness. To address this, we supplemented the DAT with an additional test that measured the number of unique words generated by participants or LLMs, a metric similar to the Alternative Uses Test⁹. Nonetheless, the Consensual Assessment Technique literature suggests that creativity assessments benefit from evaluation across multiple dimensions^{22,63}. Other aspects, such

as convergent creativity—the ability to form associations between juxtaposed concepts or words⁶⁴—are crucial for problem solving, artistic expression and communication. Future studies should consider incorporating a broader range of creativity measures to provide a more comprehensive assessment of human and LLM creative abilities.

Finally, while our investigation of prompt-engineering approaches was extensive, it represents only a subset of possible configurations. This limitation suggests caution in generalizing our findings and highlights opportunities for future research to build upon this work. Given that different creativity measures capture distinct aspects of creative ability, future studies should incorporate multiple assessment approaches, including expert evaluation methods and more complex creative tasks. Last, the rapid evolution of AI suggests that the half-life of empirical findings in this field may be relatively short. This underscores the need for continued research into the nature of AI, unpacking its black box⁶⁵ and its relationship to human thought processes⁶⁶. Future studies should explore a broader range of creativity measures beyond the DAT, investigating LLM performance on tasks requiring more complex forms of creative problem solving or artistic expression^{12–14}. Additionally, longitudinal research tracking the development of LLM creative capabilities over time could provide valuable insights into the trajectory of AI advancement in this domain. We replicated the study using later models including GPT-5, Grok 4 and Claude Opus 4 and find similar results.

Methods

The research was approved by the University of Hong Kong Human Research Ethics Committee (no. EA250313) and was carried out in accordance with the principles of the revised Declaration of Helsinki. The student data from the University of Hong Kong were collected solely for educational purposes, and reuse for research purposes in this work complies with institutional policies on the ethical use of retrospective educational materials for research purposes (no. EA250313). Consent would not have been required for educational purposes alone.

Divergent creativity score computation

While there are many creativity measures in the behavioural science literature, such as the Alternative Uses Test, all of them rely on either humans or LLMs as raters. To enhance the objectivity of evaluating creativity in humans and LLMs, we employed the DAT, a well-established and validated creativity assessment²³, developed by Olson et al.⁴⁸. In this task, participants are asked to generate ten single-word nouns that are as different from each other as possible. The creativity of each participant, reflected by the DAT score, is calculated by validating each word against a dictionary, converting the words into word embeddings and calculating the average pairwise Euclidean distances between the embeddings. Scores range from zero to over 100, with a maximum possible score of 200. The higher the score, the more creative. We believe this methodology offers a scalable, transparent and replicable comparison of creativity between humans and LLMs^{67–69}. Note that when applying descriptions, such as creativity and divergent thinking, to LLMs, we intend that LLMs simulate these rather than possess in the way humans do.

Statistical information

Our study aimed to evaluate LLMs' performance on the DAT compared with humans in divergent creativity. We compared LLMs against human participants using both first-order and second-order statistical analyses, each with parametric and non-parametric tests. For first-order statistics, we assessed the equality of means using the independent samples *t*-test (parametric) and Mann–Whitney *U* test (non-parametric). For second-order statistics, we evaluated the equality of variances using Bartlett's test (parametric) and Levene's test (non-parametric). First-order statistics provided a baseline comparison of overall performance, while second-order statistics revealed insights

into performance variability and distribution. This variability, particularly in the top percentiles, indicated capacity for exceptional creativity. For the bottom-percentile of humans, the results indicated opportunities for human–LLM collaboration and productivity improvement. The second-order analysis also helped identify tendencies towards uniformity and potential lack of divergent creativity. Examining both mean and variance allowed us to pinpoint potential outliers and exceptional performances in humans and LLMs alike. We report empirical and theoretical quantile–quantile plots, which show that some of the LLM-generated outputs do not follow normal distribution (see the quantile–quantile plots for every study in the Supplementary Information). Thus, the inclusion of both parametric and non-parametric tests ensured the robustness of our analysis, accounting for various potential data characteristics and strengthening the validity of our conclusions. No statistical methods were used to predetermine sample sizes, but our sample sizes are similar to those reported in previous publications, and no participants or data points were excluded from the analyses^{10,11,18}.

Data availability

Data for all analyses in the main manuscript and Supplementary Information are publicly available in the Open Science Framework (<https://osf.io/a9v2t>).

Code availability

Code for all analyses in the main manuscript and Supplementary Information are publicly available in the Open Science Framework (<https://osf.io/a9v2t>).

References

1. Uzzi, B. & Spiro, J. Collaboration and creativity: the small world problem. *Am. J. Sociol.* **111**, 447–504 (2005).
2. Uzzi, B., Mukherjee, S., Stringer, M. & Jones, B. Atypical combinations and scientific impact. *Science* **342**, 468–472 (2013).
3. Pomiechowska, B., Bródy, G., Téglás, E. & Kovács, Á. M. Early-emerging combinatorial thought: human infants flexibly combine kind and quantity concepts. *Proc. Natl Acad. Sci. USA* **121**, e2315149121 (2024).
4. Frank, M. R. et al. Toward understanding the impact of artificial intelligence on labor. *Proc. Natl Acad. Sci. USA* **116**, 6531–6539 (2019).
5. Grossmann, I. et al. AI and the transformation of social science research. *Science* **380**, 1108–1109 (2023).
6. Shirado, H. & Christakis, N. A. Locally noisy autonomous agents improve global human coordination in network experiments. *Nature* **545**, 370–374 (2017).
7. Shin, M., Kim, J., van Opheusden, B. & Griffiths, T. L. Superhuman artificial intelligence can improve human decision-making by increasing novelty. *Proc. Natl Acad. Sci. USA* **120**, e2214840120 (2023).
8. Rahman, H. A. The invisible cage: workers' reactivity to opaque algorithmic evaluations. *Adm. Sci. Q.* **66**, 945–988 (2021).
9. Guilford, J. P. *The Nature of Human Intelligence* (McGraw-Hill, 1967).
10. Stevenson, C., Smal, I., Baas, M., Grasman, R. & van der Maas, H. Putting GPT-3's creativity to the (alternative uses) test. In *Proc. 13th International Conference on Computational Creativity (ICCC'22)* (eds Hedblom, M. M. et al.) 164–168 (Association for Computational Creativity, 2022).
11. Haase, J. & Hanel, P. H. P. Artificial muses: generative artificial intelligence chatbots have risen to human-level creativity. *J. Creat.* **33**, 100066 (2023).
12. Chakrabarty, T., Laban, P., Agarwal, D., Muresan, S. & Wu, C.-S. Art or artifice? Large language models and the false promise of creativity. In *Proc. 2024 CHI Conference on Human Factors in Computing Systems* (Association for Computing Machinery, 2024); <https://doi.org/10.1145/3613904.3642731>

13. Tian, Y. et al. MacGyver: are large language models creative problem solvers? In *Proc. 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (eds Duh, K. et al.) (Association for Computational Linguistics, 2024); <https://doi.org/10.18653/v1/2024.naacl-long.297>
14. Doshi, A. R. & Hauser, O. P. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Sci. Adv.* **10**, eadn5290 (2024).
15. Guo, D. et al. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature* **645**, 633–638 (2025).
16. Jia, N., Luo, X., Fang, Z. & Liao, C. When and how artificial intelligence augments employee creativity. *Acad. Manag. J.* **67**, 5–32 (2024).
17. van den Broek, E., Sergeeva, A. & Huysman, M. When the machine meets the expert: an ethnography of developing AI for hiring. *MIS Q.* **45**, 1557–1580 (2021).
18. Olson, J. A., Nahas, J., Chmoulevitch, D., Cropper, S. J. & Webb, M. E. Naming unrelated words predicts creativity. *Proc. Natl Acad. Sci. USA* **118**, e2022340118 (2021).
19. Bao, L., Cao, J., Gangadharan, L., Huang, D. & Lin, C. Effects of lockdowns in shaping socioeconomic behaviors. *Proc. Natl Acad. Sci. USA* **121**, e2405934121 (2024).
20. Beketayev, K. & Runco, M. A. Scoring divergent thinking tests by computer with a semantics-based algorithm. *Eur. J. Psychol.* **12**, 210–220 (2016).
21. Brophy, D. R. Understanding, measuring, and enhancing individual creative problem-solving efforts. *Creat. Res. J.* **11**, 123–150 (1998).
22. Amabile, T. M. The social psychology of creativity: a componential conceptualization. *J. Pers. Soc. Psychol.* **45**, 357–376 (1983).
23. Long, H. & Pang, W. Rater effects in creativity assessment: a mixed methods investigation. *Think. Skills Creat.* **15**, 13–25 (2015).
24. Dumas, D., Organisciak, P. & Doherty, M. Measuring divergent thinking originality with human raters and text-mining models: a psychometric comparison of methods. *Psychol. Aesthet. Creat. Arts* <https://doi.org/10.1037/aca0000319>. (2020).
25. Beaty, R. E., Johnson, D. R., Zeitlen, D. C. & Forthmann, B. Semantic distance and the alternate uses task: recommendations for reliable automated assessment of originality. *Creat. Res. J.* **34**, 245–260 (2022).
26. Guilford, J. P. Creativity: yesterday, today and tomorrow. *J. Creat. Behav.* **1**, 3–14 (1967).
27. Wallach, M. A. & Kogan, N. A new look at the creativity-intelligence distinction. *J. Pers.* **33**, 348–369 (1965).
28. Yang, Y., Youyou, W. & Uzzi, B. Estimating the deep replicability of scientific findings using human and artificial intelligence. *Proc. Natl Acad. Sci. USA* **117**, 10762–10768 (2020).
29. Farquhar, S., Kossen, J., Kuhn, L. & Gal, Y. Detecting hallucinations in large language models using semantic entropy. *Nature* **630**, 625–630 (2024).
30. Pividori, M. Chatbots in science: what can ChatGPT do for you? *Nature* <https://doi.org/10.1038/d41586-024-02630-z> (2024).
31. Samdarshi, P. et al. Connecting the dots: evaluating abstract reasoning capabilities of LLMs using the New York Times Connections word game. In *Proc. 2024 Conference on Empirical Methods in Natural Language Processing* (eds Al-Onaizan, Y. et al.) (Association for Computational Linguistics, 2024); <https://doi.org/10.18653/v1/2024.emnlp-main.1182>
32. Todd, G., Merino, T., Earle, S. & Togelius, J. Missed connections: lateral thinking puzzles for large language models. In *Proc. 2024 IEEE Conference on Games (CoG)* 1–8 (Institute of Electrical and Electronics Engineers, 2024).
33. Cvrček, V. et al. Comparing web-crawled and traditional corpora. *Lang. Resour. Eval.* **54**, 713–745 (2020).
34. Horowitz, J. L. Bootstrap methods in econometrics. *Annu. Rev. Econ.* **11**, 193–224 (2019).
35. Efron, B. Bootstrap methods: another look at the jackknife. *Ann. Stat.* **7**, 1–26 (1979).
36. Jentsch, S. & Kersting, K. ChatGPT is fun, but it is not funny! Humor is still challenging large language models. In *Proc. 13th Workshop on Computational Approaches to Subjectivity, Sentiment, & Social Media Analysis* (eds Barnes, J. et al.) 325–340 (Association for Computational Linguistics, 2023).
37. Castillo, L., León-Villagrà, P., Chater, N. & Sanborn, A. Explaining the flaws in human random generation as local sampling with momentum. *PLoS Comput. Biol.* **20**, e1011739 (2024).
38. Angelike, T. & Musch, J. A comparative evaluation of measures to assess randomness in human-generated sequences. *Behav. Res. Methods* **56**, 7831–7848 (2024).
39. Geva, E. & Ryan, E. Linguistic and cognitive correlates of academic skills in first and second languages. *Lang. Learn.* **43**, 5–42 (1993).
40. Henrickson, L. & Meroño-Peñuela, A. Prompting meaning: a hermeneutic approach to optimising prompt engineering with ChatGPT. *AI Soc.* <https://doi.org/10.1007/s00146-023-01752-8> (2023).
41. Giray, L. Prompt engineering with ChatGPT: a guide for academic writers. *Ann. Biomed. Eng.* **51**, 2629–2633 (2023).
42. Lin, Z. How to write effective prompts for large language models. *Nat. Hum. Behav.* **8**, 611–615 (2024).
43. Aggarwal, A., Lohia, P., Nagar, S., Dey, K. & Saha, D. Black box fairness testing of machine learning models. In *Proc. 2019 27th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering* (Association for Computing Machinery, 2019); <https://doi.org/10.1145/3338906.3338937>
44. Chao, P. et al. Jailbreaking black box large language models in twenty queries. In *Proc. 2025 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)* 23–42 (Institute of Electrical and Electronics Engineers, 2025).
45. Lapid, R., Langberg, R., & Sipper, M. Open sesame! Universal black-box jailbreaking of large language models. *Appl. Sci.* **14**, 7150 (2024).
46. Chesebrough, C., Chrysikou, E. G., Holyoak, K. J., Zhang, F. & Kounios, J. Conceptual change induced by analogical reasoning sparks aha moments. *Creat. Res. J.* **35**, 499–521 (2023).
47. Beaty, R. E. & Kenett, Y. N. Associative thinking at the core of creativity. *Trends Cogn. Sci.* **27**, 671–683 (2023).
48. Te'eni, D. et al. Reciprocal human-machine learning: a theory and an instantiation for the case of message classification. *Manage. Sci.* <https://doi.org/10.1287/mnsc.2022.03518> (2023).
49. Yax, N., Anlló, H. & Palminteri, S. Studying and improving reasoning in humans and machines. *Commun. Psychol.* **2**, 51 (2024).
50. Strachan, J. W. A. et al. Testing theory of mind in large language models and humans. *Nat. Hum. Behav.* **8**, 1285–1295 (2024).
51. Bzdok, D. et al. Data science opportunities of large language models for neuroscience and biomedicine. *Neuron* **112**, 698–717 (2024).
52. Padmakumar, V. & He, H. Does writing with language models reduce content diversity? In *Proc. International Conference on Representation Learning* (Kim, B. et al.) 642–669 (ICLR, 2024).
53. Anderson, B. R., Shah, J. H. & Kreminski, M. Homogenization effects of large language models on human creative ideation. In *Proc. 16th Conference on Creativity and Cognition* (Association for Computing Machinery, 2024); <https://doi.org/10.1145/3635636.3656204>
54. Mohammadi, B. Creativity has left the chat: the price of debiasing language models. Preprint at <https://arxiv.org/abs/2406.05587> (2024).

55. Groh, M. et al. Deep learning-aided decision support for diagnosis of skin disease across skin tones. *Nat. Med.* **30**, 573–583 (2024).
 56. Marks, M. A., DeChurch, L. A., Mathieu, J. E., Panzer, F. J. & Alonso, A. Teamwork in multiteam systems. *J. Appl. Psychol.* **90**, 964–971 (2005).
 57. Vaccaro, M., Almaatouq, A. & Malone, T. When combinations of humans and AI are useful: a systematic review and meta-analysis. *Nat. Hum. Behav.* **8**, 2293–2303 (2024).
 58. Bellemare-Pepin, A. et al. Divergent creativity in humans and large language models. Preprint at <https://arxiv.org/abs/2405.13012> (2024).
 59. Chen, H. & Ding, N. Probing the ‘creativity’ of large language models: can models produce divergent semantic association? In *Findings of the Association for Computational Linguistics: EMNLP 2023* (eds Bouamor, H. et al.) (Association for Computational Linguistics, 2023); <https://doi.org/10.18653/v1/2023.findings-emnlp.858>
 60. Childs, P. et al. The creativity diamond—a framework to aid creativity. *J. Intell.* **10**, 73 (2022).
 61. Chen, L. et al. TRIZ-GPT: an LLM-augmented method for problem-solving. In *Proc. 36th International Conference on Design Theory and Methodology (DTM) V006T06A010* (American Society of Mechanical Engineers, 2024).
 62. Chen, L. et al. DesignFusion: integrating generative models for conceptual design enrichment. *J. Mech. Des.* **146**, 111703 (2024).
 63. Hennessey, B. A., Amabile, T. M. & Mueller, J. S. in *Encyclopedia of Creativity* (Elsevier, 2011); <https://doi.org/10.1016/B978-0-12-375038-9.00046-7>
 64. Cropley, A. In praise of convergent thinking. *Creat. Res. J.* **18**, 391–404 (2006).
 65. Wang, D. Presentation in self-posted facial images can expose sexual orientation: Implications for research and privacy. *J. Pers. Soc. Psychol.* **122**, 806–824 (2022).
 66. Taylor, J. E. T. & Taylor, G. W. Artificial cognition: how experimental psychology can help generate explainable artificial intelligence. *Psychon. Bull. Rev.* **28**, 454–475 (2021).
 67. Voudouris, K. et al. Direct human–AI comparison in the animal–AI environment. *Front. Psychol.* **13**, 711821 (2022).
 68. Hitsuwari, J., Ueda, Y., Yun, W. & Nomura, M. Does human–AI collaboration lead to more creative art? Aesthetic evaluation of human-made and AI-generated haiku poetry. *Comput. Hum. Behav.* **139**, 107502 (2022).
 69. Griffiths, T. L. Understanding human intelligence through human limitations. *Trends Cogn. Sci.* **24**, 873–883 (2020).
- 72574227 and T2293771). H.S. was supported by the Theme-based Research Fund provided by HKU Education Consulting (Shenzhen) Co., Ltd (grant SZRI2023-TBRF-03), the Research Grants Council of the Hong Kong Special Administrative Region, China (grant CRF-C7162-20G), and Strategic allocation 2018/19 (2c): Capacity Building for Development of ‘Business Analytics and Big Data’. B.U. was supported by the National Science Foundation through the NSF National Synthesis Center for Emergence in the Molecular and Cellular Sciences (grant MCB-2335029), Northwestern University’s Kellogg School of Management, Northwestern Institute on Complex Systems, and the Ryan Institute on Complexity. The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation.

Author contributions

D.W., D.H., H.S. and B.U. designed the research. D.W. performed the research. D.W. and H.S. analysed the data. D.W., D.H. and B.U. wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41562-025-02331-1>.

Correspondence and requests for materials should be addressed to Dawei Wang or Brian Uzzi.

Peer review information *Nature Human Behaviour* thanks Tuhin Chakrabarty, Liuqing Chen and Ken Gilhooly for their contribution to the peer review of this work. Peer reviewer reports are available.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher’s note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2025

Acknowledgements

We thank Z. Dai, K. Savani and Z. Shen for their contributions to this work. D.W. was supported by the Seed Fund for Basic Research from the University of Hong Kong (grant no. 2201101303). D.H. was supported by the National Natural Science Foundation of China (grant nos. 72503232,